LITERATURE REVIEW AND ENGINEERING BACKGROUND

Vacuum Technologies for Debris Collection and Relocation

From motor-driven suction systems to compressed-air ejectors and Line Vac arrangements

Debris pickup, conveyance, and retention

Purpose

This document reviews the physical principles, published experimental results, and practical limits of technologies that collect loose debris and relocate it through a conveying line. It progresses from electrically driven fans and vacuum pumps to fluid-driven systems, ending with compressed-air ejectors, commercial Line Vacs, and arrangements using more than one unit.

Evidence policy

Journal research provides the principal evidence. Manufacturer literature and one patent are identified separately where they establish commercial configurations or operating guidance. Reported experimental results, numerical predictions, catalog ratings, and the engineering recommendations developed here are distinguished throughout. This is a critical narrative review, not a systematic review or a certification of a particular assembly.

Reading guide

The executive synthesis states the main findings. Sections 1–3 establish the review scope and engineering background. Sections 4–7 compare mechanical generators, separation, and energy accounting. Sections 8–12 develop pneumatic systems, direct Line Vac evidence, multiple-unit arrangements, and a proposed evaluation method. The reference list includes titles and persistent DOI links.

Compilation: use `latexmk -pdf` with the supplied `.tex` file. The preamble, references, and vector diagrams are included; no external bibliography, figures, or proprietary fonts are required.

Executive synthesis

Successful debris collection requires both a pressure difference and sustained transport flow. Motor-driven impellers, positive-displacement pumps, and compressed-air ejectors can generate suction; the complete system also needs an effective pickup, a suitable transport route, and a receiver that retains particles. A blocked hose can exhibit substantial vacuum while delivering almost no useful transport flow.

The principal generator groups are centrifugal fans and multistage blowers; side-channel blowers; positive-displacement machines, including Roots, claw, screw, and liquid-ring designs; and motive-fluid ejectors. A blower-driven suction pipeline is also pneumatic conveying. Here, *compressed-air vacuum system* specifically means motive-air ejector operation. Filter bags, cyclones, and wet collectors describe separation rather than the suction mechanism.[1–3]

For electric systems, a particularly relevant laboratory study found that accumulating mortar debris substantially reduced airflow in the tested bag-prefilter vacuums, whereas cyclonic-preseparator models maintained much steadier flow. At a much larger scale, a Roots-blower installation collected coal deposits through a suction pipeline with separation ahead of the blower. These studies support selecting and testing a *complete loaded system*, rather than choosing by motor nameplate power or a maximum-vacuum number alone.[2, 4]

Particle behavior adds a second constraint. Startup from a deposited bed, continued transport, and collection from a moving surface are different processes. Published pickup correlations depend on particle characteristics and pipe size, and can show substantial disagreement for fine or irregular material. Experiments with sweeper heads also show that the time available to entrain debris changes collection performance. Consequently, a universal velocity for "small debris" cannot be defended without defining the material and the geometry.[5–8]

Compressed-air ejectors offer a compact suction source with no rotating mechanism at the pickup location. Their performance depends on motive pressure *while flowing*, entrained flow, geometry, and discharge pressure. They also consume energy through the compressed-air plant. Recent gas–solid studies demonstrate useful particle capture and transport, but their results do not establish that every ejector is more efficient than every electrically driven vacuum system.[9–11]

There is direct journal evidence for an EXAIR Line Vac conveying short wood fibers and for a Line Vac introducing sand into an experimental airflow. These establish actual particulate transport, although neither is a comparison of one versus two Line Vacs collecting mixed debris into a bag. A patent separately describes two EXAIR units on a seed-conveyance path, and the manufacturer provides guidance on distributed units for longer conveying routes. These are useful configuration precedents, with different evidentiary weight from a controlled journal experiment.[12–15]

The resulting design conclusion is conditional: **begin a compact pneumatic debris-collection investigation with one appropriately sized Line Vac and a receiver that passes the total discharge airflow with low resistance.** Establish its pickup, delivered recovery, air consumption, and behavior as the filter loads. Add a second unit when measurements reveal a transport limitation that the second unit can address. The review ends by explaining why series boosters, parallel suction generators, and separate pickup branches solve different problems, and how to compare them experimentally. This is an engineering recommendation developed from the evidence, not a published universal optimum.

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1 Scope, terminology, and review method

1.1 Research questions and application boundary

The review addresses three related questions: which mechanisms can provide suction for debris collection; what journal research demonstrates about their performance; and what evidence supports pneumatic arrangements using one or several in-line air conveyors. The application includes dry dust, sand, chips, fibers, small fragments, and mixed loose solids. Wet material and slurry systems are included where their different carrier fluid changes the preferred architecture. A specified particle size distribution, moisture condition, material density, required mass-transfer rate, hose route, and operating duty would be needed to turn this background into a final equipment selection.

The primary task is *open-flow collection*: surrounding air enters a pickup and carries material to a receiver. This differs from evacuating a sealed vessel, holding a suction cup against a nonporous object, or producing high vacuum for scientific processing. Those applications can favor high ultimate vacuum or low leakage, whereas an open debris pickup must continuously admit and transport gas. Performance claims from sealed-vessel evacuation are therefore not treated as debris-collection results.

1.2 Search and source assessment

Searches combined technology terms—*vacuum cleaner fan*, *side-channel blower*, *Roots vacuum conveying*, *claw vacuum pump*, *liquid-ring pump*, *air ejector*, *annular ejector*, *multi-stage vacuum generator*, and *Line Vac*—with *debris*, *dust*, *particle*, *pickup*, *saltation*, *collection efficiency*, *air consumption*, *series*, and *parallel*. Publisher records, DOI-linked pages, publicly available journal manuscripts, and university or government repositories were inspected. Searches included older foundational work and papers available through the stated completion date.

This was a targeted critical narrative search. It did not use a preregistered protocol, subscription-database deduplication, or a formal risk-of-bias scoring instrument, and no exhaustive publication count is claimed. Inclusion favored identifiable journal articles with a clear mechanism or collection experiment and a verifiable result. Full texts were used when accessible. For restricted articles, statements are limited to the publisher abstract or article preview; missing operating details are not reconstructed from a title or a secondary citation.

Four evidence categories are used in the synthesis:

- **Direct collection/conveying evidence:** particles or debris were actually entrained, transferred, or recovered.
- **Component evidence:** a fan, pump, or ejector was analyzed or tested, often without solids. This establishes mechanism or gas performance, not debris throughput.
- **Separation evidence:** particles were removed from a gas stream already in motion. This establishes a collector characteristic, not surface pickup.
- **Commercial/design evidence:** manufacturer ratings, application guidance, and patent disclosures establish an available arrangement, but not independent comparative performance.

Reported percentages retain their original metric. A percentage-point change in efficiency is distinguished from a relative percentage improvement. A model's signed average error is distinguished from

its absolute error. Measurements from different test rigs are not pooled into a technology league table.

1.3 A consistent vocabulary

Table 1: Terms that otherwise lead to misleading technology comparisons.

Term	Meaning used in this review
Vacuum generator	Device that removes gas from a suction region and establishes pressure below the local atmosphere.
Pneumatic conveying	Solid transport in a gas stream; the gas may be driven by an electric blower or by a motive-air ejector.
Line Vac	EXAIR's commercial in-line air conveyor; generic devices are called in-line ejector conveyors or solids eductors.
Regenerative blower	Side-channel machine with repeated momentum transfer between impeller and circulating gas.
Regenerative-air sweeper	Cleaning layout that recirculates part of its airflow through the pickup head; it need not use a side-channel blower.
Cyclonic or wet vacuum	Description of a separation arrangement, unless a specific liquid-ring generator is identified.
Multistage ejector	Integrated gas-ejector architecture with interacting stages; not automatically equivalent to several solids-through Line Vacs.

2 Physical background: how suction becomes debris collection

2.1 Pressure, airflow, and the operating point

Let p_a be local atmospheric pressure and p_s the absolute pressure at a selected suction location. Gauge pressure and positive vacuum magnitude are, respectively,

$$p_{s,g} = p_s - p_a, \qquad \Delta p_v = p_a - p_s = -p_{s,g}. \tag{1}$$

At standard sea-level atmosphere, the theoretical maximum vacuum magnitude is approximately 101.3 kPa, or 14.7 psi. A compressed-air supply of 100 psig therefore does not imply that the suction side can generate a 100 psi pressure deficit. The supply is an energy input to the ejector, and the atmosphere limits the available vacuum on an open pickup.

A useful vacuum specification gives *airflow at a stated suction pressure*, with the relevant inlet, outlet, and supply conditions. Free airflow and shutoff vacuum describe different endpoints; they are not a simultaneous operating pair. For a fixed-speed mechanical generator, the installed operating point is the intersection of the source characteristic with the total pressure demand of the connected system. The tested fan curves in the mortar-debris study provide a practical example of this relationship.[4]

For a conventional suction route, a conceptual pressure budget is

$$\Delta p_{\text{source}}(Q) = \Delta p_{\text{pickup}} + \Delta p_{\text{line}} + \Delta p_{\text{bends}} + \Delta p_{\text{solids}} + \Delta p_{\text{separator}} + \Delta p_{\text{filter}}. \tag{2}$$

Each contribution depends on operating conditions. A loaded filter changes the system curve. Additional solids require acceleration and can produce collisions, settling, and intermittent plugs. For significant gas-density changes, these terms should be evaluated with a compressible-flow model or measured directly rather than summed using one assumed density everywhere.

An ejector needs a broader operating map:

$$Q_s = F(p_s, p_{0,p}, p_d, T_p, \text{geometry}, \text{solids loading}), \tag{3}$$

where $p_{0,p}$ is motive-inlet total pressure, p_d is discharge pressure, and Q_s is suction flow at a specified reference state. A single suction curve is useful only when motive and discharge conditions are held fixed. This distinction becomes essential when a second ejector or a loaded receiving filter changes those boundaries.[9, 16]

2.2 Actual flow and standard flow are different quantities

For an approximately ideal gas,

$$\dot{m}_g = \rho Q, \qquad Q_{\text{actual}} = Q_{\text{ref}} \frac{p_{\text{ref}}}{p_{\text{actual}}} \frac{T_{\text{actual}}}{T_{\text{ref}}}. \tag{4}$$

All pressures in the conversion are absolute, and temperatures are absolute temperatures in kelvin. SCFM and standard liters per minute represent gas quantity at a declared standard condition; actual liters per minute describe volume at the measurement location. Different organizations may use different reference temperatures. A catalog's *compressed-air consumption* is also a different stream from its *induced suction flow.* An air conveyor with a stated consumption of 15 SCFM has not thereby been shown to draw 15 actual CFM through its pickup hose.

Superficial gas velocity, based on the empty cross-sectional area of a circular hose, is

$$u_g = \frac{Q_{\text{actual}}}{A} = \frac{4Q_{\text{actual}}}{\pi D^2}. \tag{5}$$

For illustration, 1.0 m^3/min of actual inlet airflow gives approximately 34.0, 14.7, and 8.49 m/s in hoses with internal diameters of 25, 38, and 50 mm, respectively. These are geometric calculations, not recommended conveying velocities or Line Vac ratings. Gas velocity through the available void area differs when solids occupy appreciable cross-sectional area. Increasing hose diameter reduces superficial velocity at fixed flow, but can also reduce friction enough to permit a different operating flow. Both effects must be considered together.

2.3 Pickup, saltation, and transport have different thresholds

Pickup is the initiation of particle motion from a surface or bed. Saltation identifies a transport condition at which particles settle from suspension in a horizontal line. A particle that is already moving can behave differently from one lodged in a rough surface or an agglomerate. For a single exposed particle, a simple drag scale is

$$F_D \sim \tfrac{1}{2} C_D \rho_g A_p (u_g - u_p)^2. \tag{6}$$

This relation explains sensitivity to relative velocity, projected area, and gas density, but does not by itself predict the detachment of cohesive dust, entangled fibers, or interlocked fragments. Contact, lift, friction, cohesion, and local flow around neighboring particles can be important.

Cabrejos and Klinzing experimentally separated pickup and saltation behavior and developed a coarse-particle pickup correlation incorporating pipe diameter. Their reported applicability was for particles above approximately 100 μm. Kalman et al. used approximately 100 measurements across 24 materials to develop three pickup regimes involving particle, gas, and pipe characteristics. These studies support using material-specific transport information rather than treating particle diameter as the sole design input.[5, 6]

Gomes and Amarante Mesquita compared published correlations with sand and alumina experiments in 50, 75, and 100 mm pipes. Their analysis found substantial disagreement; even a preferred fine-particle correlation had about 100% error for a 22 μm case. The relevant conclusion is uncertainty in extrapolation, not that all correlations fail under all conditions. A conservative design process uses a correlation within its demonstrated domain and then measures representative material in the intended route.[7]

2.4 Hose losses, bends, feed rate, and blockage

For a clean-gas screening calculation at approximately constant density, the familiar Darcy–Weisbach form is

$$\Delta p_{\mathrm{gas}} \approx \left(f_D \frac{L}{D} + \sum K \right) \frac{\rho_g u_g^2}{2}. \tag{7}$$

Here f_D is the Darcy friction factor, not the Fanning factor. Debris adds losses that this expression does not capture. An oversupplied pickup can bury the available air path and produce a dense plug, even if the source has adequate clean-air performance. Metering or admitting transport air can therefore improve transfer without increasing maximum suction.

Zhu et al. quantified bend losses using 4 mm polyethylene particles in an 80 mm positive-pressure conveying line. The solids-associated loss depended on loading and bend geometry; their fitted relationship had a reported standard deviation below 4.11% for the tested conditions. This is relevant adjacent evidence for a debris hose, but it is not a universal loss formula for corrugated suction hose, cohesive dust, or mixed fragments.[17]

Czernek et al. studied flat suction nozzles feeding a 30 mm vacuum-conveying line and observed seven gas–solid flow structures. Their pressure-loss model had a small signed mean relative error of −0.83%, yet a mean absolute error of 29.5%. The latter quantifies the substantial scatter and prevents describing the model as "accurate to 1%." The work directly supports treating intake geometry and solids flow regime as part of source sizing.[18]

2.5 Metrics that permit a fair comparison

The pickup and delivered-recovery fractions differ whenever particles remain in the hose or escape elsewhere. A high recovered mass fraction can also conceal poor fine-particle control when coarse particles dominate the total mass. For a separator with clean dilution air added between measurement locations, particle-removal efficiency must account for particle *mass or number flow*, not simply a concentration ratio. This is especially relevant to ejectors, which inject additional gas.

Table 2: Recommended performance quantities. Equations define the metrics used here; study-specific definitions must still be checked.

Quantity	Definition or observation	What it resolves
Pickup fraction	$(m_0 - m_{\text{remaining}})/m_0$	Was debris removed from the source?
Delivered recovery	m_{receiver}/m_0	Did removed debris reach the intended collection vessel?
Throughput	m_{receiver}/t	How quickly was the task completed?
Gas operating point	Q_s, p_s, p_d, and supply conditions	Does the installed system sustain transport?
Specific air use	$V_{p,\text{standard}}/m_{\text{receiver}}$	How much compressed air is used per unit mass delivered?
Specific energy	$E_{\text{total}}/m_{\text{receiver}}$	What is the energy cost at a stated system boundary?
Grade efficiency	Particle removal by size fraction	Are fine particles retained as well as coarse debris?
Stability	Interruptions, pressure fluctuations, residual line mass	Is transport reliable over the required duty?

3 System architectures and technology families

3.1 A receiver upstream of the gas mover

In a conventional vacuum collector, debris enters a hose and is separated into a receiver before the cleaned gas reaches the fan or pump. The suction source acts through the receiver and its filters. The architecture is suitable for locating a relatively heavy gas mover away from a handheld or robotic pickup. It also provides a deliberate barrier between abrasive solids and the mechanism that generates suction. Li and Meng's coal-collection system illustrates this arrangement at industrial scale.[2]

A bag used *inside* an evacuated receiver and a breathable bag placed at a conveyor's *discharge* occupy different pressure boundaries. The first is part of a suction system; the second must pass the discharge gas while retaining solids. Receiver shape, strength, sealing, filtration area, and emptying method therefore belong in the design specification. Calling either arrangement a "vacuum bag" is insufficient to determine whether it will work.

3.2 A pass-through ejector upstream of the receiver

In a solids-through air conveyor, the motive jets entrain both ambient air and debris into a passage. The combined stream exits toward a collection vessel. The ejector is both a suction source at its inlet and an injector of additional gas into the downstream line. The receiving filter must accommodate the gas that actually arrives there. This architecture differs from an ejector connected only to the clean-air port of a filtered vacuum receiver.

3.3 Which families merit comparison?

Table 3 summarizes the mechanisms. It is an architecture guide rather than a universal ranking. Specific performance curves, material compatibility, and duty limits take precedence over a broad

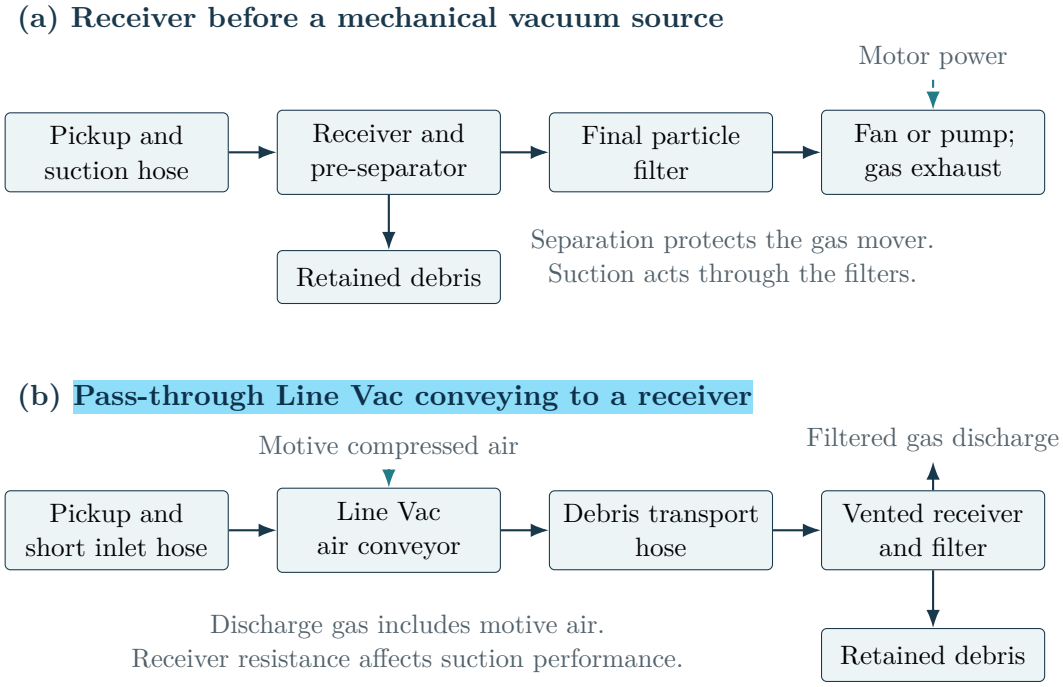


Figure 1: Two generalized collection architectures, drawn for this review. Solid arrows show process streams; dashed arrows show energy or motive-air input. The lower layout requires a suitably vented, low-resistance receiving system. Actual pressures depend on the connected network.

technology label. The review uses direct solids-handling evidence where available and explicitly identifies component-only evidence for less directly studied pump variants.

4 Electrically driven dynamic vacuum systems

4.1 Centrifugal vacuum-cleaner fans

In a motor-driven centrifugal unit, the motor rotates an impeller that accelerates gas radially. The connected flow path allows lower pressure at the inlet while the gas is discharged at a higher pressure. Compact cleaners may use high rotational speed to obtain useful suction within a small package. Their complete performance depends on the motor, impeller, diffuser, leakage paths, cooling arrangement, and the resistance of the collector. Son et al. directly analyzed this combined fan-motor flow path.[19]

Evidence and limits. For a compact unit approximately 30% smaller than its reference full-size design, Son et al. numerically optimized diffuser and guide-vane geometry. Predicted flow-path efficiency increased from 52.7% to 56.9%, a gain of *4.2 percentage points*. The manuscript states that experimental curves were withheld for commercial confidentiality. The quoted values are therefore treated here as CFD results, not measured debris-recovery efficiency or complete electrical efficiency.[19]

Engineering interpretation. A high-speed motor is only one part of the design. For an in-house collector, it is generally more informative to obtain the assembled fan's pressure–flow curve than to select a motor by electrical wattage and assume a suction capability. A flow-path improvement can be valuable, but a heavily loaded filter or poorly placed pickup may consume the resulting benefit.

Table 3: Generator families and their main design implications.

Family	How suction is generated	Main implication for debris collection
Centrifugal fan / multistage blower	Rotating impellers add gas momentum; diffuser/volute passages recover pressure.	Match fan curve to loaded hose and collector losses.
Side-channel blower	Gas repeatedly interacts with the impeller through a circulating side channel.	Useful pressure at moderate flow; monitor restriction and thermal operating limits.
Roots / rotary-lobe blower	Rotors transfer trapped gas volumes from inlet to outlet.	Supports continuous conveying flow; protect the inlet and control differential pressure.
Dry claw / screw / sliding vane	Working chambers expand at inlet and discharge gas as they contract or move to outlet.	Evaluate leakage, cooling, lubrication, and required filtration; ultimate vacuum alone is insufficient.
Liquid-ring pump	An eccentric impeller and rotating liquid ring create changing gas-chamber volumes.	Accommodates wet gas service, with operating-liquid and carryover management.
Motive-fluid ejector	A primary jet entrains secondary gas or material and the mixed flow discharges downstream.	Supply conditions, entrainment, solids passage, and discharge resistance interact.

The distinction is demonstrated by Heitbrink and Santalla-Elias. Their four tested vacuums received 15.9 kg of mortar-removal debris in 2.27 kg increments. The cyclonic-preseparator units maintained over 1.98 m^3/min, while bag-prefilter units declined from about 2.3 to as little as 0.85 m^3/min. This comparison concerns those complete machines and that debris; it does not imply that every bag collector performs poorly or that a cyclone adds no pressure loss.[4]

An electric fan system is consequently a useful baseline for recurring dry-debris collection when an appropriate power supply is available. This is a design inference, not a conclusion that one motor or fan type always minimizes energy. At mobile scale, the required electrical power, battery energy, duty cycle, mass, and thermal behavior must be considered together. Converting mains operation to battery operation changes the energy supply and operating envelope, not the suction principle.

4.2 Larger centrifugal systems and vacuum sweepers

The same basic mechanism scales to larger pickup openings and municipal litter. Singh et al. tested a mobile street cleaner over flow rates of 2,500–5,000 m^3/h and hose diameters of 0.15–0.30 m. Under their higher-flow conditions, 0.20–0.25 m hoses were preferable for the studied heavier litter, with reported nozzle velocities of 23–40 m/s. These are batch-test findings, not a transferable minimum velocity for a small debris hose. The abstract's "performance efficacy" should not be reinterpreted as a universally defined recovery efficiency.[20]

Fayzullayevich et al. combined CFD with a regenerative-air sweeper test rig. Increasing travel speed from 6 to 18 km/h reduced experimental collection from 97% to 70%; the corresponding predictions fell from 98% to 81%. Their sand tests also showed improved collection over a higher pressure-drop range. The difference between numerical and experimental travel-speed trends reinforces the need for physical pickup trials. "Regenerative air" in this study refers to recirculation at the cleaning head, not to a side-channel blower.[8]

An independent pickup-head investigation by Oruç, Açıkgöz, and Biçici used 10 kg of sand spread over a 600 × 3000 mm surface. The improved head achieved 95% sweeping efficiency at 6 km/h. This provides another example of intake geometry influencing recovery; it does not establish a measured energy reduction or a fine-particle exhaust efficiency.[21]

Engineering interpretation. For a moving robotic nozzle, debris residence time beneath the pickup and the opening-to-surface gap can matter as much as increasing source pressure. A useful development sequence is to test nozzle width, standoff, and travel speed while recording delivered mass and flow. A geometry that admits large quantities of bypass air can raise total flow while weakening entrainment where debris actually lies.

4.3 Multistage centrifugal blowers and central collection

Staging mechanical blowers can increase the pressure capability available to overcome a long network. At a common mass flow, each stage contributes work, while gas density and each stage's operating condition can change. The installed throughput response is therefore not proportional to the number of machines.

Jang and Lee instrumented automated vacuum waste-collection installations with serial centrifugal blowers. Increasing active blowers from one to seven increased the upstream pressure magnitude by about 3.7 times and inlet velocity by about 2.4 times. An auxiliary blower increased suction and energy consumption. This is direct evidence of a system-dependent staging benefit in waste collection, but it does not quantify the benefit of serial compressed-air Line Vacs.[22]

For central collection, multiple pickup branches also introduce demand variation. Opening an additional branch changes the network resistance and the available flow at other pickups. Isolation, balancing, and demand-responsive speed control are design options, but their value should be established at the intended simultaneous-use condition. A pump sized for a single open pickup cannot be assumed to support several open branches at the same velocity.

4.4 Regenerative or side-channel blowers

Side-channel blowers occupy a distinct part of the design space. Gas circulates between the impeller and a surrounding channel, receiving momentum through repeated blade passages. The resulting mechanism can produce greater pressure rise than a simple fan at a comparable modest flow, although repeated circulation also creates dissipation. The word "regenerative" describes this repeated interaction rather than recovery of otherwise free energy.[1]

Münsterjohann and Becker measured casing and blade pressures in a double-flow side-channel blower at 50 and 60 Hz rotational frequency. Across their measured operating points, flow ranged approximately 50–350 m^3/h with an inverse pressure-difference range of about 35–1 kPa. The measurements supported the circulatory-flow explanation. These are clean-air component results; the study did not establish abrasive-debris tolerance or collection efficiency.[1]

Engineering interpretation. A side-channel blower is worth considering when a filtered continuous gas stream needs more pressure capability than a selected single-stage fan can provide. The appropriate choice depends on its actual curve and permissible thermal load. A highly restricted inlet should be

evaluated against the manufacturer's operating envelope; the ability to generate vacuum does not imply unrestricted operation near shutoff.

5 Positive-displacement and liquid-ring vacuum systems

5.1 Roots and rotary-lobe blowers

Roots machines transport volumes of gas between rotating lobes and the casing. Their volumetric displacement offers a useful basis for maintaining conveying flow over a pressure range, but leakage, rotational speed, gas temperature, and permissible differential pressure remain important. They should be treated as gas machinery in a debris system, with upstream separation sized for the material loading. The presence of robust rotating lobes is not a reason to route arbitrary chips through the blower.

Li and Meng described a tunnel coal-removal system using a 110 kW electric Roots blower, cyclone separation, and secondary filtration. The paper reports coal-removal throughput above 20 t/h from the developed system and field work. The electric equipment was powered by a diesel generator, illustrating that the suction mechanism remains electrically driven even when its upstream energy source is fuel. The paper's discussion of deposits within 500 m of a tunnel entrance must not be interpreted as a validated 500 m hose-conveying distance.[2]

Engineering interpretation. Roots systems provide a credible reference architecture for continuous or larger-scale relocation: collect the solids in a receiver and allow the blower to handle conditioned gas. Scaling the cited industrial installation down by motor power alone would be inappropriate because hose diameter, solids loading, pressure ratio, and separator behavior also change.

5.2 Dry claw, screw, and sliding-vane alternatives

Claw and screw pumps create and transport working chambers with rotating elements. Sliding-vane machines use moving vanes to divide the working volume. These mechanisms can provide substantial vacuum, but their suitability for conveying depends on pumping speed *at the required pressure*, inlet filtration, gas temperature, maintenance, and whether the process tolerates lubricant or operating-fluid contact. Manufacturer application literature identifies claw and rotary-vane pumps in pneumatic suction conveying; it is used here to establish commercial application, not to prove a performance ranking.[23]

Pan et al. numerically compared conventional claw rotors with a proposed gear-claw geometry. Predicted volumetric efficiency rose from 69.5% to 77.4%, or 7.9 percentage points, while calculated inter-rotor leakage velocity fell by 21.18%. This is evidence about leakage and displacement performance. It is neither a measured electrical-energy saving nor evidence that the pump can pass abrasive particles.[24]

Li et al. developed a dry-screw pump model combining working-chamber behavior with a thermal-resistance network. The model considered leakage, heat exchange, and deformation and was checked against pumping-speed and temperature measurements. The accessible publisher abstract supports agreement and the relevance of thermal coupling, but does not provide a defensible numerical improvement to quote.[25]

Engineering interpretation. For a small open pickup, a pump that achieves a very low ultimate

pressure may still supply insufficient gas flow through the desired nozzle. Selection should therefore start with the required working flow and pressure, then compare pump variants. The reviewed evidence does not establish that claw, screw, or sliding-vane designs outperform centrifugal or Roots systems for the same mixed-debris duty. Small diaphragm and scroll pumps likewise require a flow-based justification before being considered for bulk pickup; sealed-vessel capability is insufficient.

5.3 Liquid-ring vacuum pumps

In a liquid-ring pump, an eccentric impeller drives a sealing liquid into a rotating ring. The gas spaces between blades change volume, allowing suction and compression. The operating liquid is part of the pressure-generating mechanism, unlike a wet collector that merely places water in the debris stream. Wet-gas service can favor this architecture, but the system must manage operating-liquid circulation, temperature, gas–liquid separation, and any process contamination entering the liquid.[26]

Zhang et al. experimentally studied a xanthan-gum working liquid as a means of reducing losses. At a reported inlet pressure of 60 kPa, pump efficiency increased from 34.4% with water to 43.2% at an optimum 4,500 ppm concentration; the paper reports 21.4% energy saving under that condition and 19.07% net energy-cost saving in an industrial application. These are liquid-ring operating-fluid results, not debris-collection efficiencies.[27]

A later industrial investigation by Zhang et al. reported maximum energy savings of 21–24% at an optimum 4,000 ppm concentration and implemented-system savings of 15–22% in coal-mine gas drainage. The difference between test maxima and field savings is itself useful: a component improvement and an installed-system result need not coincide. Neither paper is a recommendation to add polymer to an arbitrary pump without compatibility and service evaluation.[28]

Kaito et al. developed a particle-based liquid-ring simulation and compared it with experiments at different suction pressures. The predicted liquid-ring shapes and most casing pressures agreed with observations, although a location near 315° retained a mismatch. This provides recent mechanism-validation evidence and identifies remaining model limits. “Particles” in that numerical method are computational representations of liquid; the study did not test solid debris passing through a pump.[26]

5.4 Wet and submerged collection extensions

When the intended debris is already submerged or is deliberately transported as a slurry, a dry-air vacuum is no longer the only relevant architecture. Hydraulic suction, liquid-driven eductors, and gas–liquid airlift systems involve a liquid carrier and require liquid–solid handling at the destination. They should be assessed as a different process choice, particularly if adding water creates an undesirable waste stream.

Sadatomi, Kawahara, and Goto experimentally studied a bubble-jet airlift dredging system using 26 and 50 mm risers, each 5 m long, and four types of particles. Over submergence ratios of 0.76–0.84, their revised apparatus achieved a better particle-discharge rate than the original at the same power consumption. This is direct evidence for submerged sediment relocation; its buoyancy and multiphase-flow mechanism cannot be transferred to dry debris collection into a bag.[29]

6 Particle separation, receiver design, and filter loading

6.1 Cyclones and pre-separators

The generator creates gas motion; the collector determines how the transported debris is retained. A cyclone turns the flow so that particle inertia promotes migration toward a wall and discharge into a bin. Its usefulness depends on particle size, density, flow, geometry, and the behavior of the dust outlet. Adding a cyclone introduces resistance, so its benefit must be evaluated together with the filter loading it prevents.

Ahn et al. developed a cyclone-based vacuum-cleaner arrangement with a reported cut size of 1.5 μm at 1 m^3/min and a pressure drop of 350 mmH_2O, approximately 3.43 kPa. A cut size denotes the particle size associated with 50% grade collection under the specified conditions; it does not mean complete capture of all larger particles. The study demonstrates a compact separation design and its pressure cost, rather than a general basis for eliminating a final filter.[30]

Engineering interpretation. For chips, sand, or a broad size distribution, a pre-separator can keep much of the mass out of a finer filter. The design question is whether the additional clean resistance is outweighed by a slower increase in loaded resistance over the actual collection cycle. That comparison should include bin fill level and emptying behavior, because the receiver is part of the operating system.

6.2 Bag, wet, and final-filter performance

Trakumas et al. compared filter-bag, cyclonic, and wet collection methods. Across the tested methods, initial collection efficiencies could reach 50% at 0.35 μm and nearly 100% at 1 μm and above. Wet collection was more sensitive to airflow; cyclonic and wet collectors exhibited substantial re-entrainment, and loaded bags could also re-emit particles. These results describe particular primary collection systems and operating conditions; they do not establish the efficiency of every complete vacuum bearing one of those labels.[3]

Lin et al. examined six cordless handheld vacuum cleaners marketed with HEPA filtration. Their reported cartridge performance, with motor emissions excluded, ranged from 22% to 91% near the most penetrating particle size, with sealing and leakage affecting the assembled result. The sample was small and specific to handheld products, but the finding illustrates why a filter-media claim should not be substituted for an assembled-system leakage and filtration assessment.[31]

Engineering interpretation. A collector can recover nearly all the visible debris by mass yet release an appreciable fraction of its fine-particle number concentration. For dusty material, testing should therefore include both delivered mass and exhaust particle behavior. In addition, the same final filter that improves retention can reduce pickup flow as it loads. Filter pressure drop, bypass leakage, and recovery belong in a common test record.

6.3 A Venturi scrubber is a separator, not necessarily the suction source

A wet Venturi scrubber accelerates a particle-laden gas stream and introduces liquid to promote particle capture in droplets; a downstream stage then separates the liquid. Although its name shares a fluid-mechanical term with a vacuum ejector, the scrubber's collection result does not establish the

suction performance of a compressed-air debris pickup.

Costa et al. tested biomass-combustion particulate matter in a Venturi scrubber. Mean efficiencies were 96.6% for particles above 2.5 μm, 85.5% for 1–2.5 μm, and 66.9% below 1 μm. Their best stated condition, 262 m/s throat velocity and 2.5 L/min liquid flow, produced 99.6%, 98.9%, and 75.2%, respectively. The size dependence is central: excellent coarse-particle removal did not imply equally high submicron collection.[32]

Engineering interpretation. Wet collection may be useful where the material and disposal process tolerate water. It also creates a liquid-management task. For dry debris intended for immediate reuse or dry disposal, that change of process may outweigh a particular scrubber's capture benefit. Neither a wet scrubber nor a liquid-ring pump is interchangeable with a breathable bag simply because each can appear in a vacuum system.

7 Energy accounting and a defensible technology comparison

7.1 Useful gas power versus electricity consumption

At modest pressure differences with approximately constant gas density, a screening estimate of useful gas power is

$$P_{\text{air}} \approx \Delta p\, Q. \tag{8}$$

The pressure difference and actual flow must refer to the same operating point and a consistent measurement boundary. Multiplying a manufacturer's free-flow rating by its shutoff vacuum is invalid because those values occur at different states. Electrical input also includes motor and drive losses, aerodynamic or volumetric losses, cooling, and any continuously running auxiliaries.

For ideal isothermal compression of suction gas from absolute pressure p_s to ambient pressure p_a, a lower-bound work rate is

$$\dot{W}_{\text{iso,min}} = p_s Q_s \ln\left(\frac{p_a}{p_s}\right). \tag{9}$$

Here Q_s is actual inlet flow at p_s, and the expression assumes an ideal gas with no leakage or mechanical loss. It is a thermodynamic comparison quantity, not a prediction of motor input. It also does not include particle acceleration or all dissipative losses in a complete conveying system.

For compressed-air systems, the energy boundary must extend upstream to air production. Compressor loading and unloading, treatment, distribution losses, leaks, regulator losses, and whether the application creates additional compressor demand influence attributable electrical energy. The U.S. Department of Energy sourcebook provides the system-level basis for evaluating such end uses and considering lower-pressure alternatives where suitable. Its older example economics should not be treated as current electricity prices or universal compressor efficiencies.[11]

7.2 An illustrative compressed-air lower bound

Consider the manufacturer's nominal 14.70 SCFM motive-air consumption for a 1 in standard Line Vac at 80 psig supply.[33] For an explicit illustrative calculation, take the standard reference pressure as 101.325 kPa, take the reference and compressor-inlet temperatures as equal, and neglect humidity.

Then

$$Q_{\text{ref}} = 14.70(0.000471947) = 0.006938 \text{ m}^3/\text{s}, \tag{10}$$

$$p_{0,p} = 101.325 + 80(6.89476) = 652.91 \text{ kPa absolute}, \tag{11}$$

$$\dot{W}_{\text{iso,min}} = p_{\text{ref}} Q_{\text{ref}} \ln\left(\frac{p_{0,p}}{p_{\text{ref}}}\right) \approx 1.31 \text{ kW}. \tag{12}$$

This is the ideal isothermal power to produce that quantity of air at the stated supply pressure. Actual electrical input is greater and depends on the plant. Two units each receiving that nominal consumption correspond to twice the reference flow and an ideal lower bound of approximately 2.62 kW. These calculations do not estimate suction flow, debris throughput, or a measured compressor input, and they do not imply that a 1.31 kW electric vacuum would deliver equivalent pickup.

Engineering interpretation. Compactness at the pickup and energy efficiency at the plant are different attributes. A small ejector can be an attractive actuator while demanding appreciable upstream energy. Conversely, a long pneumatic transfer completed in a short burst can have acceptable energy per task even when its instantaneous compressed-air demand is high. The decision requires duty and delivered mass, not the absence of an electric motor at the nozzle.

7.3 Metrics and boundaries for a useful comparison

For a complete transfer task, define

$$e_{\text{task}} = \frac{\int_0^{t_f} P_{\text{attributable}}(t)\, dt}{m_{\text{delivered}}}, \qquad a_{\text{task}} = \frac{\int_0^{t_f} Q_{p,\text{ref}}(t)\, dt}{m_{\text{delivered}}}. \tag{13}$$

The first is specific energy, such as kWh/kg; the second is specific standard-air use, such as standard m^3/kg. Standard-air use becomes an energy estimate only after applying a stated, defensible air-production model or measured specific compressor energy. The integration interval should include operating steps needed for the task, such as clearing residual material, rather than terminating at an arbitrarily favorable point.

Two comparisons answer different questions. *Equal task* testing asks which system moves the required mass over the route with acceptable retention and reliability. *Equal resource* testing asks how effectively the systems use a specified electricity or compressed-air budget. Both are useful, but neither should be presented as the other. A fast system may consume more energy per kilogram; a low-energy setting may fail the required cycle time.

No retrieved study compares all these families on the same debris, nozzle, route, receiver, and duty. An honest synthesis therefore identifies mechanisms and supported results, then specifies how to close the application-specific evidence gap. The remainder of the review concentrates on compressed-air systems and, especially, the design question of one versus several Line Vacs.

Table 4: Application-led screening before detailed equipment selection. Entries are engineering interpretations of the reviewed mechanisms, not a universal efficiency ranking.

Duty characteristic	Initial candidate family	Evidence needed before selection
Repeated dry pickup with convenient electricity	Centrifugal vacuum with receiver and filtration	Loaded-system pressure–flow behavior and recovery.
Continuous filtered gas flow with appreciable line resistance	Side-channel or positive-displacement blower	Working curve, temperature limits, solids separation.
Central collection or larger solids throughput	Staged centrifugal or Roots installation	Branch demand, transport regime, receiver capacity.
Wet gas compatible with an operating liquid	Liquid-ring vacuum system	Liquid condition, separation, efficiency at working pressure.
Compact, intermittent pneumatic pickup	Ejector on a filtered receiver or solids-through air conveyor	Motive-air use, suction map, duty energy, exhaust path.
Pneumatic relocation along a demanding route	Matched in-line conveyor; possible distributed booster	Pressure along the route and measured benefit of the added unit.

8 Pneumatic vacuum systems: ejector principles and journal evidence

8.1 How motive air generates suction

A compressed-air ejector expands a primary gas stream through a nozzle, producing a high-velocity jet. Momentum transfer between that jet and a secondary stream induces suction. The streams mix and then discharge, often through a diffuser that trades velocity for pressure recovery. No rotating impeller is required within the ejector, but the upstream compressor still supplies work. At sufficiently large pressure ratios, nozzle choking, shock structures, secondary-flow restriction, and diffuser separation can influence performance; a simple incompressible Bernoulli argument does not describe the whole operating range.[9, 34]

A common gas-performance metric is the entrainment ratio

$$ER = \frac{\dot{m}_{g,s}}{\dot{m}_{g,p}}, \tag{14}$$

where the numerator is secondary *gas* mass flow and the denominator is primary motive-gas mass flow. Unless explicitly defined otherwise, conveyed solids are not included in either quantity. A manufacturer's total-flow amplification factor may instead be $(\dot{m}_{g,p} + \dot{m}_{g,s})/\dot{m}_{g,p} = 1 + ER$. Volumetric ratios agree with these mass ratios only when their reference densities agree. Consequently, entrainment ratio, total airflow amplification, vacuum magnitude, and solids throughput should not be used interchangeably.

Ejector performance must also be matched to discharge conditions. A unit that entrains freely into ambient air can behave differently when attached to a long transport hose or a restrictive filter. Increasing motive pressure may increase flow within one regime, encounter a diminishing-return region,

or move the apparatus to a different compressible-flow state. The correct engineering object is a family of operating curves with defined boundaries, not a single maximum-vacuum value.

8.2 Different pneumatic devices solve different tasks

Three arrangements require distinction.

1. **A clean-gas vacuum ejector on a receiver.** Solids enter a vessel and are retained there; the ejector withdraws filtered gas through a separate port. The gas generator does not need to pass the bulk debris, but it must overcome the receiver and filter resistance.
2. **A solids-through in-line conveyor.** Compressed-air jets act around or into an open transport passage, inducing inlet suction and carrying the combined gas–solid stream downstream. Passage size, injection geometry, abrasion, and particle shape matter directly.
3. **An air amplifier or specialized ejector.** Annular or Coanda-type arrangements may entrain large gas flows, but their suitability for solids depends on the actual geometry. Air amplification alone is not evidence of reliable mixed-debris transport.

These categories can overlap in particular commercial designs. The important distinction is the path followed by the solids. Small suction ports or internal restrictions in a conventional vacuum generator can be inappropriate for direct debris passage even when the device provides useful suction on a filtered receiver. Similarly, a multistage gas ejector should not be assumed to provide an unobstructed path through which chips can pass.

8.3 Clean-air studies: useful mechanisms, limited debris evidence

Macia et al. experimentally identified a supersonic vacuum ejector's characteristic curves and used them to predict sealed-volume evacuation time. Accounting for the system's flow characteristics reduced prediction error from 7.6% to 1.4%. This supports the use of measured characteristics in transient modeling, but a tank-evacuation result does not establish an open pickup's particle-transfer rate.[34]

Niu and Zhang compared geometry optimization with a bypass-based, integrated two-stage approach using closed-port vacuum targets of 80 and 90 kPa. The stage-based approach favored lower-vacuum operation, while geometry optimization favored stronger-vacuum conditions. Closed-port vacuum and open-port entrainment are different operating states. The findings establish a gas-ejector design tradeoff, not a result for two separate Line Vacs connected in a debris hose.[35]

Chávez-Módena et al. combined experiments and numerical analysis of a Coanda-type industrial air amplifier, varying annular gaps of 0.08, 0.15, and 0.23 mm and inlet pressure over 0.25–0.70 MPa. Increasing inlet pressure increased flow, while amplification displayed diminishing returns. The reported pressure convention should be checked before converting those values to a compressor gauge setting. This is clean-air evidence for geometry sensitivity, not a demonstration of abrasive solids passing through a commercial Line Vac.[36]

Engineering interpretation. Clean-air ejector papers are valuable for selecting measurements and explaining mechanisms. Their contribution to a debris design is incomplete until the effect of solids feed, material accumulation, and downstream separation is established. Adding stages can improve a

selected region of a gas-performance map without improving the task that matters at the pickup.

8.4 Gas–solid ejectors: what experiments demonstrate

AbdEl-hamid et al. examined gas–solid flow through ejector mixing ducts and tail sections. Adding solids reduced the measured vacuum by 7.5%, 25%, and 27% across three tested ejector configurations. The magnitude depended on geometry. This directly shows why an unloaded air test can overstate performance during conveying, while providing no basis for a single correction factor applicable to every material and ejector.[16]

Xu, Liu, and Pang combined numerical and experimental work on a powder ejector employing a double-Venturi effect. In the studied 2 in solids-eductor arrangement, a nozzle position 30 mm from the ejector outlet gave a local optimum among the evaluated positions. The phrase double-Venturi describes the internal pressure/entrainment arrangement; it is not a test of two separate EXAIR Line Vacs in series. The optimum is tied to the investigated geometry and operating conditions, not a universal spacing prescription.[37]

Ma et al. used CFD–DEM and experiments to study a Venturi ejector for dazomet granules. Their selected geometry had a 15 mm throat, a 34.5° contraction angle, and a 12° diffusion angle. Reported application precision was 99.23%, 98.46%, and 98.56% at 15, 20, and 25 g/s, respectively. Application precision is a metering/delivery metric; it should not be relabeled fine-dust capture efficiency or filter removal efficiency.[38]

Together, these studies establish that ejectors can transfer particulate material and that solids loading and geometry change performance. Their materials, diameters, and definitions differ too much to rank the designs by one quoted percentage. For irregular debris, the useful transferable lesson is to test particle passage and delivery under load rather than infer them from air-only suction.

8.5 Recent dust-capture optimization and a blower comparison

Wang, Wang, and Guo validated a gas–solid ejector model on a full-scale rig over 300–700 kPa *absolute* motive pressure. At 600 kPa absolute, approximately 499 kPa gauge at standard atmosphere, the optimized geometry raised ER from 1.22 to 1.51, a 23.8% increase. The reported downstream concentration reduction rose from 81.3% to 93.8%, or 12.5 percentage points, while total pressure drop decreased 17.1%.[9]

That concentration metric is not automatically equal to material recovered in a receiving bin or to final-filter efficiency. In a system receiving added clean motive air, concentration can also respond to dilution. A transfer test should therefore assess gas flow and particle mass flux, together with physical recovery, before treating concentration reduction as complete material capture. This is a measurement-design implication, not a claim that the cited authors failed to account for their experiment.

Venkatesan et al. compared blower-driven and air-ejector vacuum conveying using a surrogate for a difficult process powder, with fluidized and non-fluidized feeding. The accessible publisher preview reports non-fluidized saltation velocities of 5.2 m/s for the blower system and 1.85 m/s for the ejector system, together with lower ejector-system pressure drop.[10] These are findings for the tested systems, not an intrinsic reduction in the same material's transport threshold caused solely by changing the energy source. Local gas state, feed, and flow path must be reconciled before generalizing. Complete

cost and compressed-air energy inputs were not available for independent audit in the accessible material, so this review does not present a numerical life-cycle saving.

Engineering interpretation. The recent evidence supports pneumatic ejectors as serious candidates for particle handling. It does not establish a universal energy advantage over an electric vacuum. A fair comparison must use the same task, account for compressed-air production, and preserve the distinction between suction, transport, and retention.

9 Line Vac systems for debris collection and relocation

9.1 Commercial mechanism and its system implications

EXAIR's Line Vac is an in-line compressed-air conveyor with a central passage for gas and material. Motive-air injection produces suction at the inlet and drives the mixed stream toward the outlet. The open flow path distinguishes this application from a small-port ejector intended to evacuate a sealed receiver. The manufacturer's standard and heavy-duty families differ in geometry, air use, and catalog vacuum; they should be compared by model, not by the generic Line Vac name alone.[33]

For a debris-collection system, three coupled functions must succeed: ambient air must entrain material at the pickup; the combined stream must carry it through the downstream line; and the receiver must retain solids while allowing gas to escape. The compressor setting is only one input to those functions. A long suction hose before the unit can consume available inlet pressure, while a long discharge hose or loaded receiver can impose backpressure. Practical layouts therefore require both suction-side and discharge-side measurements.

9.2 Direct journal evidence involving an actual Line Vac

Short wood fibers: Fahimian et al. (2016). Fahimian, Kortschot, and Sain used a 25.4 mm internal-diameter EXAIR Line Vac to convey short wood fibers into a thermoset-processing apparatus. The selected MDF-derived fibers were approximately 0.3–0.7 mm long. At a reported 620.5 kPa (90 psi) motive setting, the paper gives 0.41 m^3/min primary air, 0.85 m^3/min combined outlet air, and approximately 11 kPa vacuum. The fiber-transfer tests used five 60 s measurements at each pressure setting. Fiber entanglement required attention to feeding.[12]

This is directly relevant evidence that a Line Vac can entrain and relocate fibrous solids. It does not establish a mixed-debris recovery percentage, and the reported flows should retain their source context because a common standard-state definition is not explicit in the cited passage. They should not be combined with values from the paper's other operating configurations to construct an unsupported performance curve.

Sand injection: Olshefski et al. (2022). Olshefski et al. used a Line Vac to introduce sand into the airflow of a particle-sampling-probe test facility. A feeder provided up to 10 g/min, and the paper described a 552 kPa supply and −29.9 kPa gauge vacuum as sufficient to feed sand without backflow.[13] The study's principal validation concerns the sampling probe; the cited pressure is an apparatus specification, not a measured Line Vac performance map. Nevertheless, the installation is a second journal precedent for real particulate transport using the commercial device.

Evidence and limits. These papers support the feasibility of conveying specific fibers and sand with a Line Vac. Neither tests collection from a floor into a breathable bag, quantifies all residual material in a long hose, or compares one unit against two. The absence of such a retrieved comparison is a limit of this search, not proof that no unpublished or inaccessible industrial trial exists.

9.3 Model-specific ratings and what they do not establish

Table 5 reproduces selected manufacturer ratings at a nominal 80 psig (5.5 bar gauge) supply. The air quantities are *compressed-air consumption*. They are not suction flow or solids capacity, and the catalog vacuum entries alone do not provide a simultaneous suction-flow operating point.[33]

Table 5: Selected EXAIR catalog data at 80 psig. Manufacturer data; no independent loaded-debris performance is implied. Negative vacuum values are gauge pressures as presented in the catalog.

Family	Model	Nominal size	Motive air	Catalog vacuum
Standard	6058	3/8 in	5.60 SCFM	−29.9 kPa
Standard	6081 / 6061	1 in	14.70 SCFM	−11.0 kPa
Heavy duty	150100	1 in	35 SCFM	−26.0 kPa
Heavy duty	150200	2 in	75 SCFM	−10.0 kPa

The smaller standard unit illustrates why vacuum magnitude alone cannot rank conveying capacity: its listed vacuum is greater than that of the larger standard unit, but its passage and gas demand differ. Conversely, a larger passage is not evidence that a particular debris bed will be picked up more effectively. A useful selection requires suction flow, discharge tolerance, and material behavior at the actual working point.

Two standard 1 in units each operating at their stated supply condition would nominally consume 29.4 SCFM together. A single 1 in heavy-duty unit is listed at 35 SCFM. This is a comparison of catalog air demand only. It does not show which arrangement gives greater recovery, throughput, or energy efficiency. If the supply cannot maintain pressure at both operating units, adding their nominal ratings does not describe the delivered operating condition.

9.4 The receiver must pass the added motive air

For steady gas flow through a solids-through system with no other gas sources or leakage, conservation of gas mass gives

$$\dot{m}_{g,\mathrm{receiver}} = \dot{m}_{g,\mathrm{pickup}} + \sum_{i=1}^{N} \dot{m}_{g,p,i}. \tag{15}$$

Standard-volume flows may be added in the same way when they share a reference state. Local actual-volume flows at different pressures or temperatures cannot simply be summed. The receiver and its final filter must be designed for the total gas arriving there, while the pickup velocity depends on the gas entering the suction inlet.

A sealed bag with no gas outlet cannot accept a continuous air stream indefinitely. Pressure rises until the operating state changes, flow finds another path, or transfer stops. A breathable bag is a filter and

has a pressure drop that varies with area, material, and loading. A vented rigid receiver can provide more controlled separation, but its vents and filters still belong in the discharge-pressure budget.

EXAIR's backpressure guidance states that Line Vacs are intended to vent to ambient and advises against discharge into a pressurized vessel or an outlet smaller than the unit.[39] This is manufacturer operating guidance, not a claim that a real hose has zero pressure loss. The available pressure margin should be established for the chosen model and route, and the receiving system should avoid unnecessary restriction.

Engineering interpretation. Adding a finer bag without measuring pressure can improve nominal retention while undermining transport. A larger effective filter area, appropriate pre-separation, shorter restrictive connections, or a different receiver may solve the observed limitation more effectively than increasing motive pressure. These are hypotheses to test at the required loading, not guaranteed improvements for every debris type.

10 Multiple Line Vacs: configurations, evidence, and limits

10.1 Several distinct meanings of multiple units

Using two ejectors does not define one architecture. Table 6 separates the main possibilities; Figure 2 illustrates two of them. The distinctions follow from gas conservation and the pressure boundaries, not from the commercial name on the device.

Table 6: Configurations using multiple pneumatic units. Effects are conditional engineering interpretations.

Arrangement	Intended function	Main interaction to verify
Units distributed in series along one material route	Add momentum downstream and support relocation over a difficult segment.	Second-unit suction changes first-unit discharge pressure; both add motive gas.
Units immediately back-to-back	Modify the combined source/conveyor behavior in a compact length.	Strongly coupled inlet/outlet states; possible diminishing returns.
Parallel ejectors on one filtered suction receiver	Increase available gas withdrawal at a common suction pressure.	Flow sharing, common supply capacity, and receiver losses; ultimate vacuum does not simply add.
Separate pickup branches with separate units	Serve several collection locations.	Compressor demand and any shared receiver interact; pickups do not automatically reinforce one another.
Integrated multistage ejector	Extend or improve a designed gas-evacuation operating region.	Internal stage and valve behavior; solids passage may be unsuitable.

10.2 Manufacturer guidance for serial Line Vacs

EXAIR's application note on one versus two Line Vacs states that a second unit can improve conveying speed or distance but will not simply double performance. It suggests locating a booster beyond the midpoint of a longer route or at a bend, depending on the application, and describes diminishing gains from successive back-to-back units.[15] This is useful practical guidance from the manufacturer. The

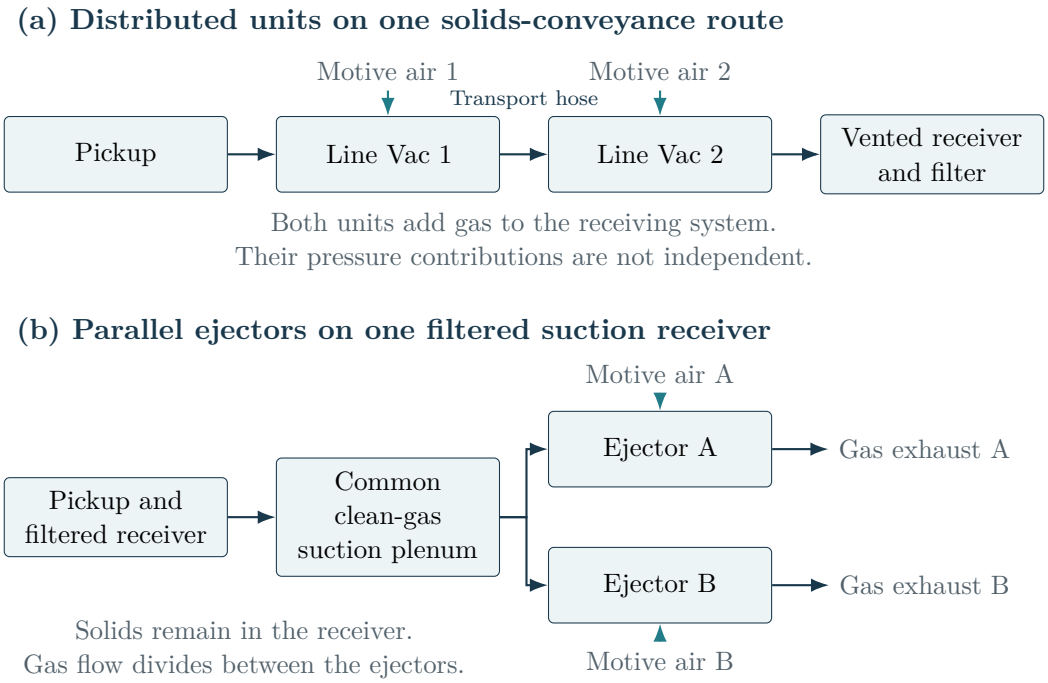


Figure 2: Original schematic comparisons. In (a), debris passes through both units. In (b), solids are separated before the gas divides. Geometry is conceptual, not a validated assembly or prescribed spacing.

note does not supply a controlled dataset with material properties, full pressure/flow measurements, uncertainty, and recovery for one versus two units.

Engineering interpretation. That guidance is best used to define candidate layouts for testing. It should not be converted into a universal optimum distance, an additive vacuum rule, or an assured percentage increase in throughput. The relevant position is where a unit can improve the pressure and momentum conditions of the actual route without imposing unacceptable air use or receiving-system load.

10.3 A documented two-Line-Vac material-handling arrangement

A U.S. patent application by Blickhan et al. describes two EXAIR Model 6063 units, identified as LV2 and LV3, on a seed-conveyance path. The disclosed route moves material from a collection chute through tubing toward a cyclone/diffuser and counting equipment. Controlled air operation can be repeated to clear remaining seeds or debris.[14]

This is relevant because it documents a concrete multi-unit arrangement for relocation and clearing of particulate material. Its evidentiary limit is equally important: the patent is a design disclosure, not a peer-reviewed test comparing one and two Line Vacs. It provides no basis for quoting a measured vacuum increase, collection efficiency, energy reduction, or universally preferred spacing. The use of seeds also leaves questions about abrasion, cohesive fines, and mixed-fragment blockage unresolved.

10.4 Why vacuum values cannot simply be added

In a mechanical compressor train, stages can be analyzed at a common mass throughput, with each stage's gas state affecting the next. In a series of solids-through ejectors, each unit introduces additional primary gas. The mass throughput therefore changes along the route. The second unit's inlet pressure is coupled to the first unit's discharge, and the common compressed-air supply can change both motive conditions.

A downstream ejector may reduce the first unit's discharge pressure and help the upstream unit entrain more effectively. It also increases the total gas that must be transported and discharged. Depending on the pressure map and receiver, either effect may dominate. Therefore the statement that the second unit always increases backpressure is as unjustified as the statement that it always doubles suction. The coupled response has to be measured or calculated with appropriate ejector characteristics.

Likewise, two parallel ejectors on a common plenum can provide greater combined gas withdrawal at a shared suction pressure if both operate effectively and the supply supports them. Their maximum vacuum magnitudes do not add. If the units instead serve separate pickup hoses and share only a downstream collection vessel, they do not constitute parallel suction sources for a single pickup. They still interact through the shared air supply and receiver.

Engineering interpretation. For a proposed multiple-Line-Vac system, first draw the gas and solids paths and label the pressure boundaries. Identify which ports actually share a plenum, which units pass debris, and which receiver sees the combined exhaust. This simple accounting determines whether the intended improvement is additional pickup capacity, downstream transport assistance, or coverage of another collection location.

11 A practical research program for one versus two Line Vacs

11.1 Hypotheses and controlled configurations

The most useful unresolved question is application-specific: does a second unit deliver a worthwhile improvement in recovered mass, task time, or reliable distance for the intended debris, after accounting for air demand and receiver behavior? A small controlled experiment can answer this more directly than extrapolating a sealed-vessel vacuum rating.

Begin with a single Line Vac and a reproducible debris batch. Record particle-size range, shape, density where available, moisture state, and the presence of fibers or agglomerates. Fix pickup geometry, standoff, hose material, internal diameter, total route length, bends, elevation change, receiver, and filter condition. These variables can dominate the response.

When a second unit is absent, substitute an appropriate plain transport section so that route length is preserved. Document any remaining local-geometry difference rather than claiming perfect equivalence. For distributed placement, pressure measurements can identify a likely location; manufacturer guidance supplies a starting candidate, not proof of an optimum.

Run two clearly labeled comparisons if resources allow. First, maintain the prescribed flowing motive pressure at each active unit; this measures the benefit obtained by allowing additional air demand. Second, impose a common total standard-air budget and compare admissible settings within each unit's

Table 7: Proposed experimental configurations. This is a study design developed for this review, not a published standard.

Test	Configuration	Question addressed
A	One unit near the pickup; full transport route.	What does a well-matched baseline accomplish?
B	Same route with a second unit at a measured or hypothesized downstream limitation.	Does distributed injection improve transfer?
C	Two units back-to-back, if compact staging is a candidate.	Does close coupling help more or less than distribution?
D	One alternative higher-capacity unit, if available.	Is one larger unit preferable to two smaller ones?

operating envelope; this tests allocation efficiency. Equal total air use and equal motive pressure at every active unit cannot generally be imposed simultaneously across one- and two-unit configurations.

11.2 Measurements that isolate the controlling limitation

At a minimum, acquire the following synchronized measurements:

- Motive pressure *at each unit during flow*, plus standard motive-air consumption for each branch. Tank pressure or a regulator's no-flow setting is insufficient.
- Static pressure at the pickup, immediately before and after each ejector, in the receiver, and across the final filter. Use protected pressure taps that do not accumulate debris.
- Suction airflow at a known local pressure and temperature. A protected or appropriately placed calibrated flow measurement is preferable to inferring flow from compressor consumption.
- Initial debris mass, mass remaining at the pickup, mass delivered to the receiver, and residual mass recovered from the hose and components after the run.
- Task time, interruptions, plugging events, and any material escaping collection. For dusty material, include an appropriate exhaust particle measurement.

A mass balance,

$$m_0 = m_{\text{remaining}} + m_{\text{receiver}} + m_{\text{line}} + m_{\text{other}}, \tag{16}$$

helps distinguish poor pickup from good pickup followed by poor relocation. The last term represents accounted spills, sampling, escape, and measurement residual; it should not be silently assigned to successful capture. If the receiver is weighed during flow, jet forces, pressure forces, and hose tension can bias the scale. A settled, no-flow weighing provides a useful check.

11.3 Loaded operation, repeatability, and energy

Test both a documented clean-filter state and a reproducible loaded state. Changing the bag after every trial may hide the degradation that matters in routine use. Randomize or alternate the configuration order to reduce drift from compressor temperature, material conditioning, and filter loading. At least five repeat runs per condition are a reasonable initial design choice, not a statistical guarantee; increase replication when observed variability prevents a useful comparison.

Report the mean and variability of recovery, throughput, specific air use, and interruption rate, alongside instrument uncertainty. Include startup and any required clearing period in a predefined task interval. A configuration that leaves material in the hose until a long purge should not be credited with a short transfer time based only on the disappearance of debris at the pickup.

Where electric-energy comparison is important, measure compressor energy over a boundary that accounts for storage. Depleting a compressed-air receiver during one trial and recharging it after measurement would undercount the task's energy. Either restore comparable initial and final stored-air states or account for the change. For a shared compressor plant, distinguish marginal task energy from an allocation of the whole plant's idle consumption and explain the choice.

Engineering interpretation. Adopt the second unit when it produces a useful improvement in the required task and the additional air, noise, maintenance exposure, and receiver demand are acceptable. If its benefit disappears as the filter loads, first investigate the receiving path. If the one-unit system already meets the duty with lower specific air use, the experiment has also supplied a useful answer.

12 Concluding synthesis: pneumatic opportunities and remaining evidence gaps

The literature establishes compressed-air ejectors as practical devices for creating suction and conveying particles. Gas-only research explains entrainment and stage behavior; gas–solid experiments demonstrate sensitivity to material loading and geometry; direct Line Vac studies document transfer of short fibers and sand. Those bodies of evidence answer complementary questions. None should be substituted for a measured recovery and energy result from the complete debris-collection task.

For multiple Line Vacs, the retrieved evidence is less complete. Manufacturer guidance and a two-unit seed-handling patent establish plausible layouts, while controlled journal evidence directly comparing one and two branded units on the same debris route was not identified. The most consequential research gaps are therefore:

- Pressure–flow–solids maps for in-line conveyors under realistic discharge resistance.
- Controlled comparisons of one unit, distributed boosters, and a single larger unit at both equal task and equal air budget.
- Receiver and fine-filter loading behavior when multiple units add motive gas.
- Mixed debris, moisture, irregular fragments, fiber bridging, abrasion, and repeated start–stop operation.
- Complete mass recovery and compressor-energy measurements with uncertainty and a stated storage boundary.

For an initial pneumatic debris collector, one correctly matched Line Vac, a short and effective pickup path, a suitable downstream hose, and a low-resistance filtered receiver provide a defensible baseline. This configuration should be assessed using delivered recovery and air per unit mass, not maximum vacuum alone. A second Line Vac is then a targeted means of changing transport conditions, with a benefit that must exceed its additional air demand and receiving-system burden for the intended duty.

The engineering value of pneumatic vacuum technology lies in its ability to place a compact, mechanically simple suction-and-conveyance device where material is collected. Its successful use depends on treating the compressor, ejector, solids route, and receiver as one system. The proposed measurements make that system concrete and allow the decision to use one or several Line Vacs to rest on demonstrated collection and relocation performance.

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